MA 301

Test 3

Spring - 2013

Problem	points	out of
1		10
2		10
3		10
Score		30

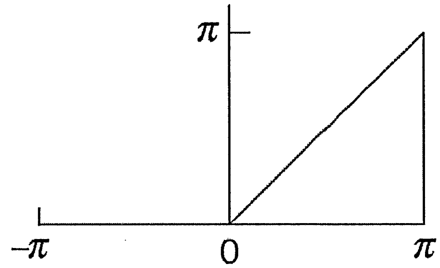
Name: SOLUTIONS

HR: ______________

Signature: ______________

Formula Sheet
The Fourier series of a function, $f(x)$, with period, $2L$, is given by $$f(x) = a_0 + \sum_{n=1}^{\infty}\left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right)\right]$$ where $$a_0 = \frac{1}{2L}\int_{-L}^{L} f(x)\,dx$$ $$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)dx$$ $$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)dx$$

Problem 1: Determine the Fourier Series for the function depicted in the graph below. Assume it periodically extends along the x-axis.



$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx$$

$$= \frac{1}{2\pi}\left[\int_{-\pi}^{0} 0\,dx + \int_{0}^{\pi} x\,dx\right]$$

$$= \frac{1}{2\pi}\left[0 + \frac{1}{2}x^2\Big|_0^{\pi}\right] = \frac{1}{2\pi}\left(\frac{\pi^2}{2}\right) = \frac{\pi}{4}$$

$$f(x) = \begin{cases} 0 & -\pi \le x \le 0 \\ x & 0 \le x \le \pi \end{cases}$$

$f(x) = f(x+2\pi)$

f is neither odd nor even

$P = 2\pi \implies L = \pi$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)\,dx$$

$$= \frac{1}{\pi}\int_0^{\pi} x\cos(nx)\,dx \qquad \left\{\begin{matrix} u = x & v = \frac{1}{n}\sin(nx) \\ du = dx & dv = \cos(nx)dx \end{matrix}\right\}$$

$$= \frac{1}{\pi}\left[\frac{x}{n}\sin(nx) - \frac{1}{n}\int \sin(nx)\,dx\right]_0^{\pi}$$

$$= \frac{1}{\pi}\left[\frac{x}{n}\sin(nx) + \frac{1}{n^2}\cos(nx)\right]_0^{\pi} = \frac{1}{\pi}\left[\left(\frac{\pi}{n}\sin(n\pi) + \frac{1}{n^2}\cos(n\pi)\right) - \left(0 + \frac{1}{n^2}\cos(0)\right)\right]$$

$$= \frac{1}{\pi}\left[0 + \frac{(-1)^n}{n^2} - \frac{1}{n^2}\right] = \frac{(-1)^n - 1}{\pi n^2}$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(nx)\,dx = \frac{1}{\pi}\int_0^{\pi} x\sin(nx)\,dx \qquad \left\{\begin{matrix} u = x & v = -\frac{1}{n}\cos(nx) \\ du = dx & dv = \sin(nx)dx \end{matrix}\right\}$$

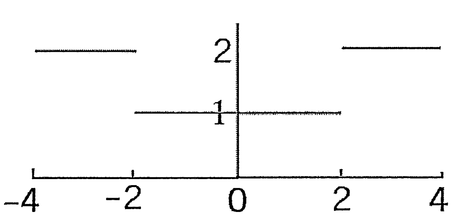
$$= \frac{1}{\pi}\left[-\frac{x}{n}\cos(nx) + \frac{1}{n}\int \cos(nx)\,dx\right]_0^{\pi} = \frac{1}{\pi}\left[-\frac{x}{n}\cos(nx) + \frac{1}{n^2}\sin(nx)\right]_0^{\pi}$$

$$= \frac{1}{\pi}\left[\left(-\frac{\pi}{n}\cos(n\pi) + \frac{1}{n^2}\sin(n\pi)\right) - \left(0 + \frac{1}{n^2}\sin(0)\right)\right]$$

$$= \frac{1}{\pi}\left[-\frac{\pi(-1)^n}{n}\right] = -\frac{(-1)^n}{n} = \frac{(-1)^{n+1}}{n}$$

$$f(x) = \frac{\pi}{4} + \sum_{n=1}^{\infty}\left[\frac{(-1)^n - 1}{\pi n^2}\cos(nx) + \frac{(-1)^{n+1}}{n}\sin(nx)\right]$$

Problem 2: Determine the Fourier Series for the function depicted in the graph below. Assume it periodically extends along the x-axis.



$$a_0 = \frac{1}{4}\int_0^4 f(x)\,dx = \frac{1}{4}\left[\int_0^2 1\,dx + \int_2^4 2\,dx\right]$$

$$= \frac{1}{4}\left[2+4\right] = \frac{6}{4} = \frac{3}{2}$$

$$f(x) = \begin{cases} 2 & -4 \le x \le -2 \\ 1 & -2 \le x \le 2 \\ 2 & 2 \le x \le 4 \end{cases}$$

$f(x) = f(x+8)$, $P = 8 \Rightarrow L = 4$

f is even $\Rightarrow b_n = 0$

$$a_n = \frac{2}{4}\int_0^4 f(x)\cos\left(\frac{n\pi x}{4}\right)dx$$

$$= \frac{1}{2}\left[\int_0^2 1\cos\left(\frac{n\pi x}{4}\right)dx + \int_2^4 2\cos\left(\frac{n\pi x}{4}\right)dx\right]$$

$$= \frac{1}{2}\left[\frac{4}{n\pi}\sin\left(\frac{n\pi x}{4}\right)\Big|_0^2 + \frac{8}{n\pi}\sin\left(\frac{n\pi x}{4}\right)\Big|_2^4\right]$$

$$= \frac{1}{2}\left[\frac{4}{n\pi}\left(\sin\left(\frac{n\pi}{2}\right) - \sin(0)\right) + \frac{8}{n\pi}\left(\sin(n\pi) - \sin\left(\frac{n\pi}{2}\right)\right)\right]$$

$$= \frac{1}{2}\left[\frac{4}{n\pi}\sin\left(\frac{n\pi}{2}\right) - \frac{8}{n\pi}\sin\left(\frac{n\pi}{2}\right)\right]$$

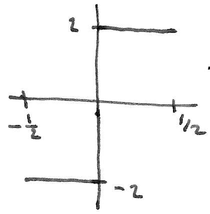
$$= \left(\frac{2}{n\pi} - \frac{4}{n\pi}\right)\sin\left(\frac{n\pi}{2}\right) = -\frac{2}{n\pi}\sin\left(\frac{n\pi}{2}\right)$$

$$\Rightarrow \quad f(x) = \frac{3}{2} + \sum_{n=1}^{\infty} -\frac{2}{n\pi}\sin\left(\frac{n\pi}{2}\right)\cos\left(\frac{n\pi x}{4}\right)$$

Problem 3: For each function $f(x)$ below, circle the Fourier Coefficients that must be zero. Let $g(x)$ be an even function of period 1 and $h(x)$ an odd function with period 1.

$$f(x) = \begin{cases} 0 & -2 \le x \le 0 \\ 1 & 0 \le x \le 1 \\ 0 & 1 \le x \le 2 \end{cases} \quad \text{and} \quad f(x) = f(x+4) \qquad a_n \quad b_n$$

NEITHER ODD NOR EVEN



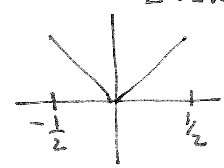
$$f(x) = \begin{cases} -2 & -\frac{1}{2} \le x \le 0 \\ 2 & 0 \le x \le \frac{1}{2} \end{cases} \quad \text{and} \quad f(x) = f(x+1) \qquad a_n \quad b_n$$

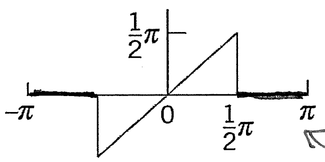
ODD

← ODD

$$f(x) = |x| \quad \text{and} \quad f(x) = f(x+1) \qquad a_n \quad b_n$$

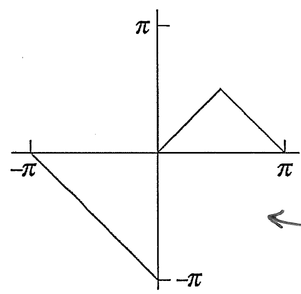
EVEN





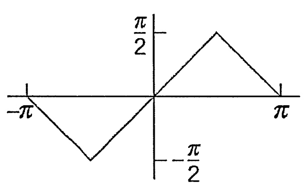
and $f(x) = f(x+2\pi)$ $\quad a_n \quad b_n$

← ODD



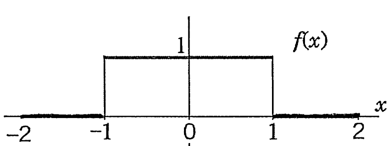
and $f(x) = f(x+2\pi)$ $\quad a_n \quad b_n$

← NEITHER



and $f(x) = f(x+2\pi)$ $\quad a_n \quad b_n$

← ODD



and $f(x) = f(x+2)$ $\quad a_n \quad b_n$

← EVEN

$$f(x) = \overbrace{\underbrace{2g(x)}_{E}}^{E} \quad \text{and} \quad f(x) = f(x+1) \qquad a_n \quad b_n$$

EVEN

$$f(x) = \underbrace{\underbrace{g(x)}_{E} \cdot \underbrace{h(x)}_{O}}_{O} \quad \text{and} \quad f(x) = f(x+1) \qquad a_n \quad b_n$$

ODD

$$f(x) = \underbrace{\cos((h(x))}_{E} \quad \text{and} \quad f(x) = f(x+1) \qquad a_n \quad b_n$$

Cosine is EVEN